Cardiovascular System (CARD) · NPLEX-P1-CARD-13

CV-13 Cardiovascular Biochemistry and Nutrition

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-13-01-C01				
NPLEX-P1-CARD-13-01-C02				
NPLEX-P1-CARD-13-01-C03				
NPLEX-P1-CARD-13-01-C04				
NPLEX-P1-CARD-13-01-C05				
NPLEX-P1-CARD-13-01-C06				
NPLEX-P1-CARD-13-01-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-13-01-C01-BD1

Build the lipoprotein table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C01 Activity: NPLEX-P1-CARD-13-01-C01-BD1 Date: _____

Without looking at your notes, build a table of the five lipoproteins in order of decreasing size, with one row each for chylomicron, VLDL, IDL, LDL, and HDL. Use the columns made by, key apolipoproteins, and main role, and fill every cell. Beside the table write what happens to a particle's size, density, and cholesterol content as triglyceride is removed, and explain why density runs in the order you wrote. Then write one line each for apo B-48, apo B-100, apo C-II, apo E, and apo A-I saying what that apolipoprotein does. Finish by naming the two enzymes that esterify cholesterol, and write where each one works and which particle or compartment it acts in.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-13-01-C01-BD2

Three broken steps in lipid transport

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C01 Activity: NPLEX-P1-CARD-13-01-C01-BD2 Date: _____

Three patients each lack one component. Patient 1 lacks apo C-II. Patient 2 lacks ABCA1. Patient 3 lacks apo E function because both apo E alleles are E2. For each patient write which particle accumulates in plasma, which step stopped, and why the particles upstream cannot move on. Then write which plasma lipid rises most in each one and whether plasma left overnight in the cold would show a creamy top layer, uniform turbidity, or neither, and explain each appearance from the particle involved. Finish by explaining why a high fasting triglyceride usually comes with a low HDL, naming the transfer protein and the enzyme that produce that pairing and what each one does to the HDL particle.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-13-01-C01-RT01

Rebuild the exogenous lipoprotein pathway

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C01 Activity: NPLEX-P1-CARD-13-01-C01-RT01 Date: _____

Rebuild the six-step exogenous pathway from the enterocyte to the liver, one step per line and in order. Beside the packaging step write the apolipoprotein used and the transfer protein that loads the core. Beside the transport step write the route the particle takes to reach the blood and why it cannot enter a blood capillary directly. Beside the enzyme step write which apolipoprotein activates the enzyme, where the enzyme sits, and what it releases. Mark on your drawing the step that fails in lipoprotein lipase deficiency, and write how long after a meal a normal plasma sample should be free of these particles. Underneath, rebuild the endogenous pathway in the same way from VLDL secretion to LDL clearance, naming the enzyme that makes IDL, the enzyme that makes LDL, and the apolipoprotein LDL depends on.

Rebuild from memory: Exogenous lipoprotein pathway

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-13-01-C01-Q01

Reverse cholesterol transport fails at its first step

A 28-year-old man is evaluated for numbness and weakness of both hands. Examination shows enlarged orange-yellow tonsils and reduced sensation in a glove distribution. Laboratory studies show total cholesterol of 110 mg/dL, HDL cholesterol of 3 mg/dL, and triglyceride of 210 mg/dL. Macrophages in his bone marrow are filled with cholesteryl ester. Which of the following steps is most likely defective in this patient?

A. Transfer of free cholesterol and phospholipid from the cell membrane to lipid-poor apolipoprotein A-I

B. Esterification of free cholesterol on the surface of HDL by lecithin-cholesterol acyltransferase

C. Hydrolysis of chylomicron triglyceride by lipoprotein lipase on capillary endothelium

D. Hepatic uptake of remnant particles through apolipoprotein E

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-13-01-C02-BD1

Build the Fredrickson table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C02 Activity: NPLEX-P1-CARD-13-01-C02-BD1 Date: _____

On blank paper build a table of the primary dyslipidemias with one row each for type I, type IIa, type IIb, type III, type IV, and type V. Use the columns defect, inheritance where the notes gave one, raised particle, which lipids are raised, and consequences, and fill every cell. Then write the four desirable adult lipid values from the notes with their units: total cholesterol, LDL cholesterol, HDL cholesterol in men and in women, and fasting triglyceride. Finish by listing the four secondary causes that must be excluded before any pattern is called familial.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-13-01-C02-BD2

Name the type from the numbers

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C02 Activity: NPLEX-P1-CARD-13-01-C02-BD2 Date: _____

Four patients have fasting lipid studies. Patient 1 is a 9-year-old with abdominal pain, triglyceride of 2,800 mg/dL, cholesterol of 260 mg/dL, and a creamy layer over clear plasma after refrigeration. Patient 2 is a 41-year-old with a body mass index of 34, triglyceride of 450 mg/dL, cholesterol of 210 mg/dL, HDL of 31 mg/dL, and uniformly cloudy plasma. Patient 3 is a 38-year-old with cholesterol of 360 mg/dL, triglyceride of 340 mg/dL, and yellow palmar creases. Patient 4 is a 12-year-old with cholesterol of 390 mg/dL, triglyceride of 120 mg/dL, thickening of both Achilles tendons, and a father who had a myocardial infarction at 40. For each patient write the Fredrickson type, the particle that is accumulating, the molecular defect, and the single finding that settled it. Then explain the mechanism behind the abdominal pain in patient 1 and why that patient's atherosclerotic risk is not greatly raised, and explain why the liver in patient 4 keeps making cholesterol and secreting VLDL even though plasma cholesterol is high.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-13-01-C02-CP01

Type I against type IIa against type III

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Type I	Type IIa	Type III
Molecular defect and inheritance			
Particle that accumulates			
Cholesterol and triglyceride pattern			
Appearance of plasma left overnight in the cold			
Skin or eye findings			
Atherosclerotic risk and why it is high or low			
What else must be present for the pattern to appear			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-13-01-C03-BD1

Cholesterol synthesis and the four controls on the reductase

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C03 Activity: NPLEX-P1-CARD-13-01-C03-BD1 Date: _____

Write out cholesterol synthesis from acetyl-CoA to cholesterol, naming every intermediate the notes gave in order. Mark the rate-limiting step, name its enzyme, say which membrane that enzyme sits in, and write how many NADPH the reaction uses. Name the branch point intermediate and list the three non-sterol products made from it. Then write the four levels at which the rate-limiting enzyme is controlled, and for each one name the signal and say whether activity goes up or down. Finish by writing which hormones raise its activity or expression and which lower them.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-13-01-C03-BD2

Follow one LDL particle into the cell and out again

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C03 Activity: NPLEX-P1-CARD-13-01-C03-BD2 Date: _____

Follow one LDL particle from the plasma into a hepatocyte. Write the apolipoprotein that binds, the receptor tail motif that gathers the receptor into a coated pit, the coat protein, the protein that pinches the vesicle off, what the falling pH in the endosome does to the receptor and the particle, where each one goes next, the enzyme that frees cholesterol in the lysosome, and the two proteins that carry that cholesterol out. Beside each of three steps write the disorder named in the notes that blocks it. Then write the three things the released cholesterol does as a feedback signal. Finish by explaining why hypothyroidism raises plasma LDL, and why macrophage scavenger receptor uptake of oxidized LDL keeps going until the cell is full of cholesteryl ester.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-13-01-C03-RT01

Rebuild SREBP activation

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C03 Activity: NPLEX-P1-CARD-13-01-C03-RT01 Date: _____

Rebuild the seven-step sequence that runs from falling membrane cholesterol to restored cell sterol, one step per line and in order. Name the escort protein, the protein it lets go of, the compartment the complex travels to, and the two proteases in the order they cut. Write which part of the transcription factor enters the nucleus and the two genes it turns on. Then draw the same system when cell sterol is high and write what holds the complex in place and what happens to the two genes. Underneath, write which SREBP controls fatty acid synthesis genes and what signal activates it.

Rebuild from memory: SREBP activation when cell sterol is low

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-13-01-C03-Q02

Why one child responds poorly

A 10-year-old girl has a total cholesterol of 680 mg/dL, a normal triglyceride, thickened Achilles tendons, and yellow plaques on both eyelids. Genetic testing shows two null alleles of the LDL receptor gene. She is given an inhibitor of HMG-CoA reductase, and her plasma LDL falls far less than it does in her heterozygous father. Which of the following best explains the small response in this patient?

A. Hepatic cholesterol synthesis is already maximally suppressed by the high plasma LDL concentration

B. Apolipoprotein B-100 on her LDL particles is structurally abnormal and cannot be recognized

C. Most of the plasma effect of the drug comes from SREBP-2 driven production of new LDL receptors, which her cells cannot make

D. The drug blocks the branch of the pathway that makes ubiquinone rather than the branch that makes sterols

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-13-01-C04-BD1

Cardiac fuels and the carnitine shuttle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C04 Activity: NPLEX-P1-CARD-13-01-C04-BD1 Date: _____

Write the four fuels the heart uses. For each one write how it enters oxidation, naming the enzyme or pathway, and the state in which its share of ATP production rises. Write what fraction of resting ATP comes from fatty acids, how they arrive at the myocyte, and which transporter carries them in. Then draw the carnitine shuttle across both mitochondrial membranes, labelling CPT-I, the translocase, and CPT-II, and write what each one does and which chain lengths of fatty acid do not need it. Write the molecule that inhibits CPT-I, the enzyme that makes it, and the two signals that raise and lower that enzyme. Finish by writing why the liver makes ketone bodies but cannot use them, and what phosphocreatine and creatine kinase do for the myocyte.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-13-01-C04-BD2

Work out the oxygen cost of each fuel

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C04 Activity: NPLEX-P1-CARD-13-01-C04-BD2 Date: _____

Using 2.5 ATP per NADH and 1.5 per FADH2, calculate the ATP yield and the O2 consumed for one glucose and for one palmitate, show your working line by line, and convert each result to ATP per oxygen atom. State which fuel gives more ATP per molecule and which gives more ATP per oxygen atom, and write the approximate percentage difference. Then explain in three or four sentences why that difference matters to a heart whose coronary flow is limited but not absent. Finish by writing the glucose-fatty acid cycle in both directions: how high glucose and insulin reduce fatty acid entry into mitochondria, naming the molecule and the transport step involved, and how active fatty acid oxidation shuts down glucose use, naming the three metabolites and the two enzymes they act on.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-13-01-C04-RT01

Rebuild energy failure in ischemia

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C04 Activity: NPLEX-P1-CARD-13-01-C04-RT01 Date: _____

Rebuild the eight-step sequence from coronary occlusion to irreversible injury, one step per line and in order. Beside the relevant steps write the ATP yield of anaerobic glycolysis per glucose and per glucose unit from glycogen, the enzyme that regenerates the NAD+ it needs, the approximate time at which contraction fails, and the approximate time at which injury becomes irreversible. Mark the step at which acidosis begins to limit glycolysis and name the enzyme it inhibits. Then write, in order, what happens to the Na+/K+-ATPase, to extracellular K+, to intracellular Na+, to the Na+/Ca2+ exchanger, to cytosolic Ca2+, and to relaxation. Finish by naming the protein released once the membrane is irreversibly damaged and the substance whose appearance in coronary sinus blood marks ischemia.

Rebuild from memory: From coronary occlusion to energy failure

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-13-01-C04-PR01

Predict what happens when the carnitine shuttle fails

An infant has carnitine palmitoyltransferase II deficiency and has been fasting for 14 hours. Predict each item below, write the direction of the change, and give the step or molecule that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to long-chain fatty acyl-CoA in the cytosol of the cardiac myocyte		
What happens to the rate of beta-oxidation in the mitochondrial matrix		
What happens to acetyl-CoA and NADH inside the mitochondrion, and therefore to pyruvate dehydrogenase activity		
What happens to the myocyte's ability to make ATP during the fast, and what the ventricular wall looks like over time		
What happens if the same infant is given a fat source made of medium-chain fatty acids, and why		
What happens to cardiac fuel use instead if the infant is given glucose, naming the molecule that rises and the transport step it blocks		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-13-01-C05-BD1

Thiamine pyrophosphate and the enzymes that need it

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C05 Activity: NPLEX-P1-CARD-13-01-C05-BD1 Date: _____

Write where thiamine is absorbed, the enzyme that converts it to its coenzyme form, and why deficiency can appear within weeks. Write which part of the coenzyme attacks the substrate, which carbon it attacks, and what is released, then say in one sentence why that chemistry is needed for both oxidative decarboxylation and transketolase. Build a table of the four enzymes that require thiamine pyrophosphate with the columns reaction catalysed and effect of deficiency, and fill every cell. Beside the dehydrogenase rows write the four other coenzymes those complexes need and the vitamin each comes from. Finish by writing the laboratory test the notes describe and what result indicates deficiency.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-13-01-C05-BD2

Explain every finding in wet beriberi

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C05 Activity: NPLEX-P1-CARD-13-01-C05-BD2 Date: _____

Rebuild the six-step mechanism of wet beriberi in order, from the enzyme block to the dilated failing myocardium. Then write each of these findings and beside it the step that produces it: warm extremities, bounding pulse with a wide pulse pressure, tachycardia, raised jugular venous pressure with leg edema, an enlarged heart on radiograph, and a raised lactate with normal oxygen saturation. Explain in two or three sentences why lactate is high when oxygenation is normal. Write the name of the fulminant form. Then write the four clinical settings in which deficiency develops, and explain why giving a glucose load to someone with marginal stores can bring on symptoms. Finish by writing the one laboratory or examination clue that separates each of severe anemia, thyrotoxicosis, and an arteriovenous fistula from wet beriberi, and the two examination findings that separate ordinary low-output failure from all of them.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-13-01-C05-CA01

Breathlessness and warm hands in a malnourished man

A 52-year-old man is brought to the emergency department because of 3 weeks of worsening breathlessness and swelling of both legs. He has drunk about a litre of vodka daily for 12 years and has eaten very little in that time. Temperature is 36.8 degrees C (98.2 degrees F), pulse is 118/min, respirations are 24/min, and blood pressure is 134/48 mm Hg. His hands and feet are warm, the pulse is bounding, the jugular venous pressure is raised, and there is pitting edema to both knees with crackles at both lung bases. Laboratory studies show hemoglobin of 13.1 g/dL, thyroid-stimulating hormone within the reference range, lactate of 6.2 mmol/L (reference below about 2), arterial oxygen saturation of 97% on room air, and normal troponin. A chest radiograph shows an enlarged cardiac silhouette.

1. Recognize. Name the nutritional deficiency and the form of heart failure it has produced, and write the two findings in this presentation that tell you cardiac output is high rather than low.

2. Explain. Write the enzymes that lose their coenzyme, and explain how one biochemical block produces both the low systemic vascular resistance and the failing myocardium.

3. Differentiate. Write how you would separate this from severe anemia, thyrotoxicosis, an arteriovenous fistula, and ordinary low-output heart failure, naming the finding in this patient that argues against each one.

4. Investigate. Write which laboratory test from the notes would confirm the deficiency, what it measures, and what result indicates deficiency, and name the other test that explains his raised lactate.

5. Decide. State why his lactate is raised when his oxygen saturation is normal, and name the fulminant form of this condition that his lactate and blood pressure should make you watch for.

6. Justify. Justify your answer from the chemistry of the coenzyme: write which reaction type it makes possible, which substrate accumulates behind the block, and why the nervous system and the heart are affected before other tissues.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-13-01-C05-Q03

A high lactate with normal oxygenation

A 46-year-old woman who has had persistent vomiting for 5 weeks after bariatric surgery is admitted with breathlessness and leg edema. Pulse is 116/min and blood pressure is 128/46 mm Hg. Her extremities are warm and the pulse is bounding. Arterial oxygen saturation is 98% on room air, and venous lactate is 5.8 mmol/L (reference below about 2). Addition of the active coenzyme to a sample of her red cells raises transketolase activity sharply. Which of the following enzyme pairs has lost its coenzyme in this patient?

A. Carnitine palmitoyltransferase I and carnitine palmitoyltransferase II

B. Methionine synthase and cystathionine beta-synthase

C. Lysyl oxidase and prolyl hydroxylase

D. Pyruvate dehydrogenase and alpha-ketoglutarate dehydrogenase

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-13-01-C06-BD1

Draw the methionine cycle and its two exits

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C06 Activity: NPLEX-P1-CARD-13-01-C06-BD1 Date: _____

Draw the methionine cycle from memory with methionine, S-adenosylmethionine, S-adenosylhomocysteine, and homocysteine in a loop. Label the step that uses ATP, the step that donates a methyl group, and the hydrolysis step. Draw both exits for homocysteine: name the remethylating enzyme with its vitamin cofactor and its methyl donor, name the enzyme that makes that methyl donor, name the liver enzyme that remethylates by a second route, and name the two enzymes of transsulfuration with the vitamin they share and the amino acid they produce. Mark which exit removes homocysteine permanently. Then write what the methylfolate trap is, which deficiency causes it, and the one laboratory value that separates vitamin B12 deficiency from folate deficiency, naming the other enzyme that explains it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-13-01-C06-BD2

Two ways micronutrients let a vessel fail

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C06 Activity: NPLEX-P1-CARD-13-01-C06-BD2 Date: _____

Write the four mechanisms by which raised homocysteine injures a blood vessel, naming the pathway it impairs in the anticoagulant system and what it does to nitric oxide. Then write the normal fasting range for homocysteine and say how a mildly raised value should be read and why. Next, a 7-month-old boy has kinky pale hair, hypotonia, low serum copper and ceruloplasmin, and tortuous elongated arteries. A separate 60-year-old man has bleeding gums, perifollicular hemorrhages, bruising, and poor wound healing. For each one write the enzyme that has failed, the metal or vitamin it needs, which structural protein cannot be finished, which bond or residue is missing, and how that explains the vascular finding. Write where in the cell each of those two enzymes acts. Finish by writing the gene and inheritance of the infant's disease, two acquired causes of the same deficiency, and the disease caused by the opposite transport defect.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-13-01-C06-CP01

Sorting out a raised homocysteine

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Cause	Step in the pathway that is affected	Methylmalonic acid	Other findings that point to it
Folate deficiency			
Vitamin B12 deficiency			
Vitamin B6 deficiency			
Cystathionine beta-synthase deficiency			
MTHFR C677T variant			
Chronic kidney disease			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-13-01-C07-BD1

Map each electrolyte onto the action potential and the tracing

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C07 Activity: NPLEX-P1-CARD-13-01-C07-BD1 Date: _____

Draw a ventricular myocyte action potential and label phase 4, phase 0, phase 2, and phase 3. Beside each phase write the ion and current that produce it and the part of the ECG it corresponds to. Then build a table with one row each for hyperkalemia, hypokalemia, hypocalcemia, hypercalcemia, and hypomagnesemia, using the columns change in the action potential, ECG findings, and main rhythm risk, and fill every cell. Beside the table write the adult reference range for potassium, total calcium, ionized calcium, and magnesium with their units, the correction to total calcium for a low albumin, the artefact that falsely raises a potassium result, and the reason a normal plasma magnesium can hide a real deficiency.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-13-01-C07-BD2

Why one potassium goes high and another will not come up

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-13 Cardiovascular Biochemistry and Nutrition
Competency: NPLEX-P1-CARD-13-01-C07 Activity: NPLEX-P1-CARD-13-01-C07-BD2 Date: _____

Take a potassium that is rising from 4.5 to 8.5 mEq/L. Write the first change on the tracing, the current responsible, and why that current responds before anything else. Then write what has to happen to the resting membrane before the QRS widens, which channels are inactivated, and the order in which the P wave, the PR interval, and the QRS change. Separately, a woman on a loop diuretic has a potassium of 2.8 mEq/L that does not rise despite replacement, and her QT interval is long. Write which other electrolyte you would measure and why its loss keeps the potassium low, naming the renal channel involved, and write the two other consequences of that same deficiency given in the notes. Finish by explaining how a long QT becomes torsades de pointes, naming the depolarization type, the channel that drives it, and the beat sequence that often precedes it, then write why a low potassium increases cardiac glycoside toxicity.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-13-01-C07-PR01

Predict the tracing in kidney failure

A man with advanced kidney failure has a potassium of 7.4 mEq/L (reference 3.5 to 5.0 mEq/L), a total calcium of 6.8 mg/dL (reference 8.5 to 10.5 mg/dL), and an albumin of 4.0 g/dL. Predict each item below, write the direction of the change, and give the current, phase, or channel that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to phase 3 of the ventricular action potential and to the shape of the T wave		
What happens to the length of phase 2 and to the ST segment		
What happens to the QT interval, and which of the two electrolyte changes is driving that		
What happens to the resting membrane potential and to the fraction of fast Na+ channels available		
What happens to the P wave, the PR interval, and the QRS duration as potassium rises further		
What the tracing looks like immediately before cardiac arrest and which two rhythms this patient is at risk of		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

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Next retrieval date: _____

Board-style question NPLEX-P1-CARD-13-01-C07-Q04

A potassium that will not correct

A 58-year-old woman taking a thiazide diuretic and a proton pump inhibitor is admitted after an episode of collapse. Her plasma potassium is 2.7 mEq/L (reference 3.5 to 5.0 mEq/L) and remains at 2.8 mEq/L after two days of potassium replacement. The electrocardiogram shows flat T waves, prominent U waves, and a corrected QT interval of 0.52 s, and telemetry captures a short run of polymorphic ventricular tachycardia. Which of the following best explains why her potassium has not corrected?

A. Reduced parathyroid hormone release is holding her ionized calcium down and prolonging phase 2

B. Magnesium depletion has removed its normal block of renal ROMK channels, so potassium continues to be lost in the urine

C. A raised extracellular potassium concentration is increasing IKr and speeding phase 3 repolarization

D. Alkalosis has increased albumin binding of calcium, which raises the threshold of excitable membranes

Why did you pick it?